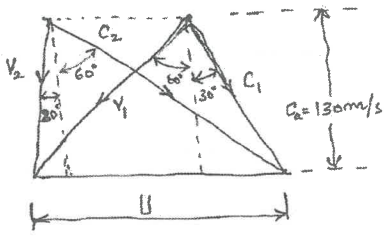
USN. | | | | | | | | | | [24MTAPT101]

# FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

## AIRCRAFT PROPULSION

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:**
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

| | | Questions | CO's | Blooms Level | Marks |
|---|---|---|---|---|---|
| **1.** | **a.** | Draw a neat block diagram of a turbofan engine and label its parts. Mention an important difference between a turbofan and a turbojet engine. What is the engine design parameter that makes a turbofan engine essentially perform like a turbojet engine. What is the range of the parameter. | **CO1** | **L3** | **10** |
| | **b.** | Velocity triangles at mean radius (35 cm) of an axial compressor rotor is given in the Fig.1b. The axial velocity component is uniformly 130 m/s at inlet and outlet. Calculate the shaft speed. What is the power required to drive the compressor if the mass flow rate is 14kg/s.   Fig.1b | **CO2** | **L3** | **10** |
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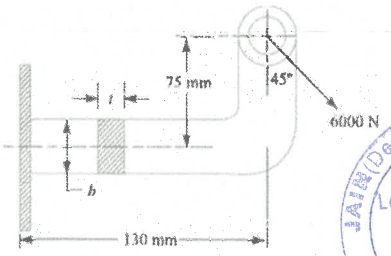
| **2.** | **a.** | A mild steel bracket as shown in figure is subjected to a pull of 6000N acting at 45° to its horizontal axis. The bracket has a rectangular section whose depth is twice the thickness. Find the cross sectional dimensions of the bracket, if the permissible stress in the material of the bracket is limited to 60MPa.   | **CO3** | **L3** | **07** |

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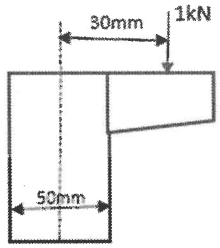
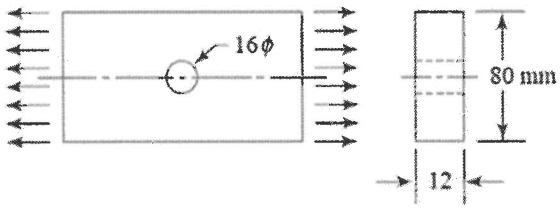
| | **b.** | A vertical column of 50mm diameter is subjected to a vertical load of 1 kN acing eccentrically at a distance of 30mm from the axis. Calculate the maximum stress in the column and locate it.   | **CO3** | **L3** | **07** |
| | **c.** | A rectangular plate of 80 mm wide and 12 mm thick with a transverse hole of 16 mm diameter is subjected to a tensile load of 20kN as shown in figure. Taking stress concentration in to account, calculate the maximum stress induced.   | **CO3** | **L3** | **06** |
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| **3.** | **a.** | Given a turbine blade with a length L = 0.5m, Young's modulus E = $2.1 \times 10^{11}$ Pa, mass per unit length of blade m = 0.5 kg/m and blade thickness t=0.01m. Calculate the fundamental natural frequency of the blade assuming it behaves like a simple beam. | **CO4** | **L3** | **10** |
| | **b.** | Construct a Campbell diagram and analyze its significance in identifying critical speeds in rotating machinery such as aircraft engine components. Illustrate with an example how the natural frequency and rotational speed interact to predict resonance conditions. | **CO4** | **L2** | **10** |
| | | **OR** | | | |
| **4.** | **a.** | Compare various analytical techniques for vibration assessment in aircraft engines. | **CO4** | **L2** | **10** |
| | **b.** | A blade in a turbine rotates at 10000 rpm. The blade has a length of 0.5m and density of blade material is 7800kg/m$^3$. Calculate the centrifugal stiffening effect on the natural frequency of the blade due to rotation. | **CO4** | **L3** | **06** |
| | **c.** | A turbo machine blade experiences resonance due to a matching of its natural frequency with the frequency of excitation caused by the rotating flow. The natural frequency of the blade is affected by | **CO4** | **L2** | **04** |

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| | | centrifugal stiffening, which increases the frequency by 8%.<br>I. If the operating speed of the machine increases by 20%, explain how this change affects the resonance behaviour of the blade.<br>II. Suggest a method to avoid resonance at the modified operating speed, considering both centrifugal stiffening and potential blade twist. | | | |
| | | | | | |
| **5.** | **a.** | Draw a neat sketch of a simple hydro-mechanical fuel control system containing four functional components and label the parts. Write one sentence on each of the functional component. | **CO5** | **L2** | **10** |
| | **b.** | Draw the revised diagram of the fuel control system if a speed control is added to the simple fuel control system mentioned in 5a. When do you recommend speed control? | **CO5** | **L2** | **10** |
| | | **OR** | | | |
| **6.** | **a.** | Draw sketches to illustrate normal and any two additional thrust vectoring nozzle function concepts of a modern military aircraft engine. What is the nominal range angle of nozzle vectoring from the normal (un-vectored) thrust axis of the aircraft for pitching. | **CO5** | **L2** | **10** |
| | **b.** | What are the benefits or advantages of variable area convergent-divergent exhaust nozzle for a supersonic airplane. Explain them briefly. What is the additional benefit of exit area control. | **CO5** | **L2** | **10** |
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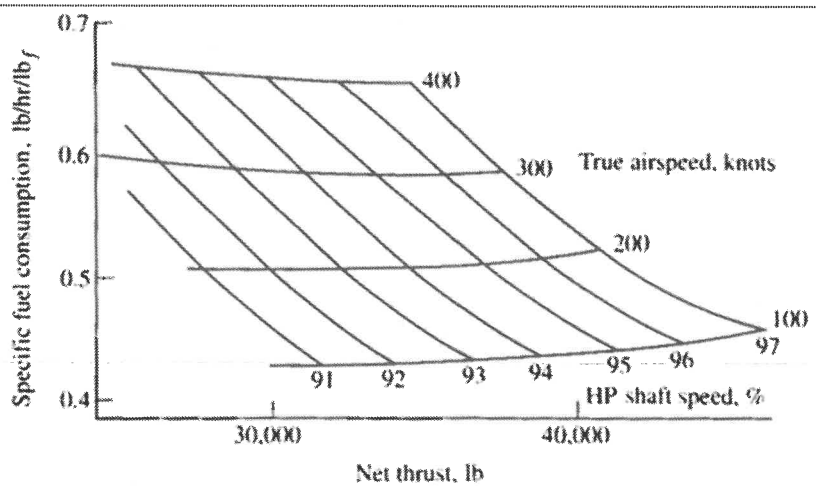
| **7.** | **a.** | Compare level of sound from various sources, under categories such as aircraft related, automotive traffic and domestic. | **CO6** | **L2** | **10** |
| | **b.** | Sketch a typical noise field from a jet spreading in still air. Show the lines representing equal sound levels. | **CO6** | **L3** | **10** |
| | | **OR** | | | |
| **8.** | **a.** |  <br>**Fig.8a1- Sea level** | **CO6** | **L3** | **10** |

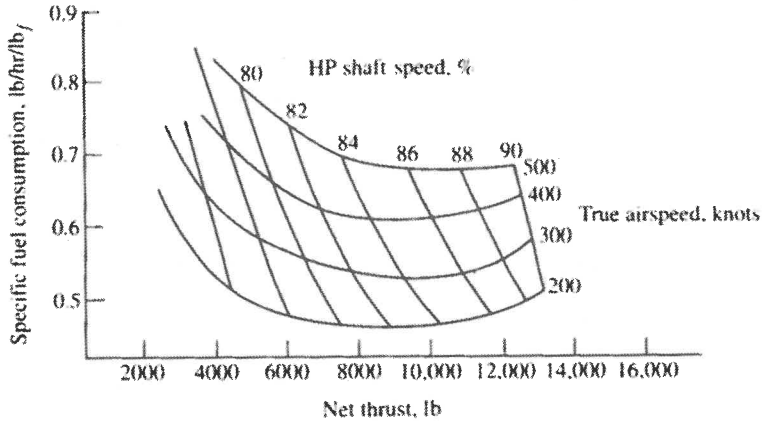



**Fig.8a2- 30,000 Feet**

Figures 8a1 and 8a2 shows the specific fuel consumption of a turbofan engine plotted against thrust as a function of high-pressure turbine shaft speed, and true airspeed (1 knot = 0.5144 m/s). At 30,000 ft altitude the standard-day temperature is 228.7 K, for which the speed of sound is 303.1 m/s. Thus, a true airspeed of 500 knots means a flight Mach number of 500 X 0.5144/303.1 = 0.848.

(i) Compare the Figs. 8a1 and 8a2 and write your comments regarding the variation of net thrust and specific fuel consumption.

(ii) What are the variables on which the net thrust depends on?

(iii) What are the variables on which the specific fuel consumption depends on?

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| | **b.** | Write the important steps to develop non-dimensional thrust and non-dimensional specific fuel consumption in terms of independent variables of flight Mach number and engine rotational tip speed Mach number. Consider the case of a typical turbofan engine. Mention the assumptions made. | **CO6** | **L2** | **10** |